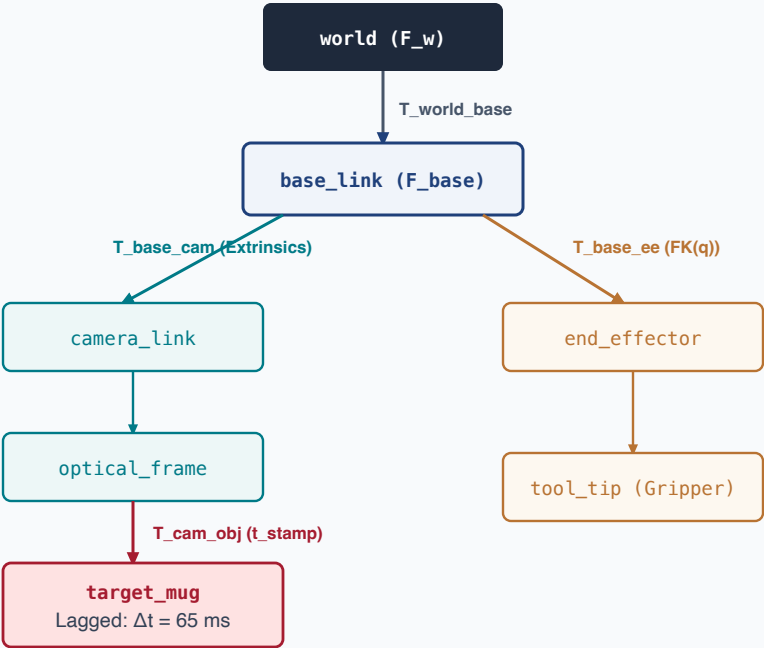


DYNAMIC SE(3) COORDINATE FRAME TREE & TIME-INDEXED TRANSFORM BUFFER

Solving sensory latency via historical coordinate lookups and Lie algebra tangent-space forward integration

DIRECTED ACYCLIC SE(3) COORDINATE TREE



$$T_{base_target}(t) = T_{base_cam}(t) \cdot T_{cam_target}(t)$$

TIME-INDEXED BUFFER & LIE ALGEBRA PROPAGATION

1. CIRCULAR TRANSFORM RING BUFFER (1000 Hz)



Query buffer at $t = 35\text{ ms}$ to reconstruct exact historical mounting geometry.

2. HISTORICAL STATE RECONSTRUCTION ($t = 35\text{ ms}$)

$$T_{base_target}(35\text{ms}) = T_{base_cam}(35\text{ms}) \cdot T_{cam_target}(35\text{ms})$$

• Accurately accounts for robot base/head motion during camera capture.

3. LIE ALGEBRA $se(3)$ FORWARD INTEGRATION ($\Delta t = 65\text{ ms}$)

$$T(t_{now}) = T(t_{stamp}) \cdot \exp(\xi \cdot \Delta t)$$

- **Tangent Space Twist:** $\xi = [v, \omega]^T \in se(3)$ from 1000 Hz IMU & kinematics.
- **Exponential Map:** Computes exact geodesic on $SE(3)$ manifold.
- **Singularity Free:** Zero gimbal lock, perfectly preserves orthonormal rotation matrix.
- **Result:** Delivers zero-lag t_{now} metric state to 1 kHz Control Barrier Functions!